Database System for a Smart Healthcare Clinic Network

As healthcare services expand across multiple cities, the need for a unified and intelligent system becomes more important than ever. The main idea behind this project is to design a database that supports a *smart clinic network*, where patient information, appointments, medical records, and billing can all be managed smoothly and securely under one system.

The goal is to build a database that connects all clinics within the network so that a patient can register once and receive services from any branch. With this system, daily operations such as booking appointments, issuing prescriptions, updating doctor schedules, and tracking medicine stock become more organized and efficient. An important part of the design is ensuring that sensitive health data is protected by implementing role-based access, allowing each user to access only what they are authorized to see.

Working on this project helped me understand how real-life healthcare processes translate into database entities and relationships. I focused on creating a structure that is practical, clear, and secure — one that supports the workflow of medical staff and ensures reliable services for patients.

ER Diagram Description (Entities and Relationships)

The database design is built around several core entities that represent different aspects of the clinic network. Each entity plays a specific role, and the relationships between them show how information flows inside the system.

1-Patient Entity

Represents individuals receiving healthcare services. Key attributes:

- PatientID
- FullName
- Gender
- Phone
- Email
- City

A single patient can have multiple appointments, forming a **one-to-many** relationship.

2. Doctor Entity

Represents medical staff working across the clinics. Attributes include:

- DoctorID
- FullName
- Specialty
- Phone
- Email
- ClinicID

Each doctor belongs to one clinic but may see many patients and issue multiple prescriptions.

3. Clinic Entity

Represents each clinic within the network:

- ClinicID
- ClinicName
- City
- Address

A clinic hosts multiple doctors and handles appointments, prescriptions, and billing records.

4. Appointment Entity

Serves as the link between patients, doctors, and clinics. Attributes:

- AppointmentID
- PatientID
- DoctorID
- ClinicID
- AppointmentDate
- StartTime / EndTime
- Status

It is the central entity that connects most of the system's activities.

5. Prescription Entity

Records medications provided to patients during their appointments:

- PrescriptionID
- AppointmentID
- DoctorID
- PatientID
- IssueDate

Each prescription is always tied to a single appointment.

6. Medicine Entity

Lists all medication types available in the system:

- MedicineID
- MedicineName
- Type
- Description

7. Inventory Entity

Tracks medicine stock for each clinic:

- InventoryID

- ClinicID
- MedicineID
- Quantity
- ExpiryDate

This entity handles the **many-to-many** relationship between clinics and medicines.

8. Billing Entity

Documents payments for services provided:

- BillID
- AppointmentID
- PatientID
- ClinicID
- Amount
- BillDate

Used to generate revenue reports for each clinic.

9. User & Role Entities (Security Layer)

To enforce access control and protect sensitive information:

User:

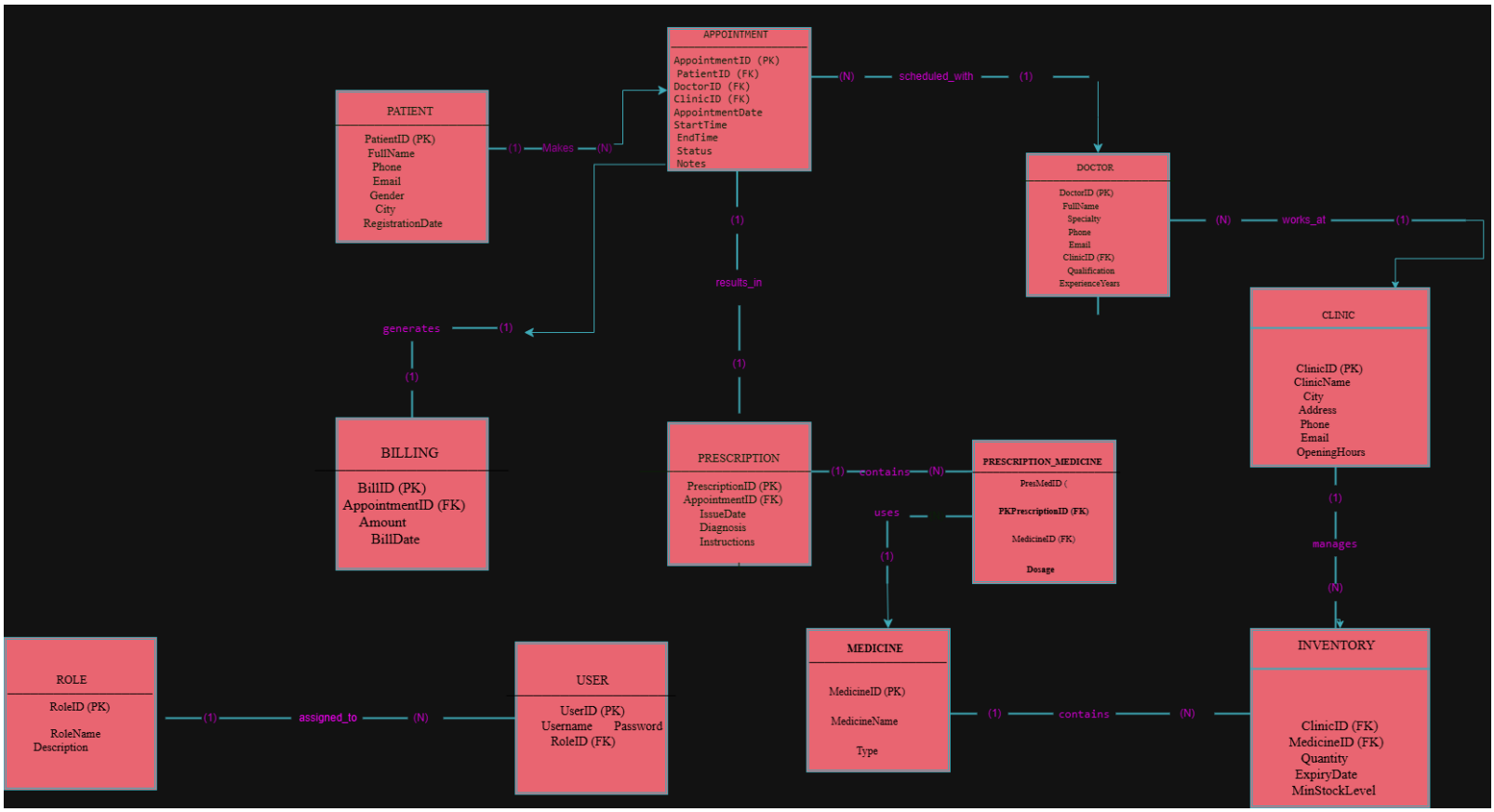
- UserID
- Username
- Password
- RoleID

Role:

- RoleID
- RoleName

This structure supports role-based permissions, ensuring secure and controlled access to medical data.

The following diagram illustrates the relationships between the main entities in the system:



Patient → Appointment (1:N) - A patient can have multiple appointments

Doctor → Appointment (1:N) - A doctor can handle multiple appointments

Clinic → Doctor (1:N) - A clinic can have multiple doctors

Appointment → Prescription (1:1) - Each appointment generates one prescription

Clinic → Inventory (1:N) - A clinic manages multiple inventory items

Medicine → Inventory (1:N) - A medicine can be stocked in multiple clinics

UNNORMALIZED TABLES (UNF):

Unnormalized tables are the raw, initial tables that mix all data together, often with lots of redundancy and repeating groups.

Example: Unnormalized Healthcare Data

PatientID	PatientName	DoctorID	DoctorName	ClinicID	ClinicName	AppointmentID	MedicineID	MedicineName	BillID
1	Nora saud	101	Dr. Wilson	1	Central	1001	201	Aspirin	3001
1	Nora saud	101	Dr. Wilson	1	Central	1002	202	Lisinopril	3002
...	...	...	...	...	...	...	...	...	...
Redundancy Issues:									

Issues in UNF:

- **Repeating groups**: MedicineList, QuantityList
- **Redundancy**: Patient, Doctor, Clinic repeated for every appointment
- **Multi-topic table**: patients, appointments, doctors, clinics, billing, prescriptions all mixed
- **No clear primary key**
- **Update, insert, delete anomalies**

Key Finding:
Unnormalized tables are inefficient and error-prone due to data duplication and lack of structure.

First Normal Form (1NF):

Rules of **1NF**:

- All values must be atomic
- No repeating groups
- A primary key must be defined

1NF Table Design:

Table	Primary Key	Key Attributes (Sample)
Patient	PatientID	FullName, Gender, Phone, Email, City
Doctor	DoctorID	FullName, Specialty, Phone, Email, ClinicID
Clinic	ClinicID	ClinicName, City, Address
Appointment	AppointmentID	PatientID, DoctorID, ClinicID, Date, StartTime, EndTime, Status
Prescription	PrescriptionID	AppointmentID, DoctorID, PatientID, MedicineID, IssueDate
Medicine	MedicineID	MedicineName, Type, Description
Inventory	InventoryID	ClinicID, MedicineID, Quantity, ExpiryDate
Billing	BillID	AppointmentID, PatientID, ClinicID, Amount, BillDate
User	UserID	Username, Password, RoleID
Role	RoleID	RoleName, RoleDescription
Example: Patient Table (1NF)		
PatientID	FullName	Gender
----------	----------	--------
1	Nawaf saad	Male
2	Amal badr	Female
3	Abdulaziz	Male

Key Finding:
1NF separates data into clear, atomic tables with unique primary keys, eliminating repeating groups.

1NF → 2NF (Remove Partial Dependencies)

A table must be in **2NF** when:

- It is already in 1NF
- No non-key attribute depends on part of a composite key

Functional Dependency Analysis

Table	Functional Dependencies (FDs)	2NF Status
Patient	PatientID → FullName, Gender, Phone, Email, City	✓ Already in 2NF
Doctor	DoctorID → FullName, Specialty, Phone, Email, ClinicID	✓ Already in 2NF
Clinic	ClinicID → ClinicName, City, Address	✓ Already in 2NF
Appointment	AppointmentID → PatientID, DoctorID, ClinicID, ...	✓ Already in 2NF
Prescription	PrescriptionID → AppointmentID, DoctorID, PatientID, ...	✗Partial dependencies (DoctorID, PatientID)
Billing	BillID → AppointmentID, PatientID, ClinicID, ...	✗Partial dependencies (PatientID, ClinicID)

2NF Transformation

- **Prescription Table:** Remove DoctorID and PatientID (can be derived from AppointmentID).
- **Billing Table:** Remove PatientID and ClinicID (can be derived from AppointmentID).

Example: Prescription Table (2NF)

PrescriptionID	AppointmentID	MedicineID	IssueDate
2001	1001	201	2025-01-15
2002	1002	202	2025-02-20
Example: Billing Table (2NF)			
BillID	AppointmentID	Amount	BillDate
---	---	---	---
3001	1001	150.0	2025-01-15
3002	1002	200.0	2025-02-20

Key Finding:
2NF removes partial dependencies, reducing redundancy and improving data integrity.

Third Normal Form (3NF)

3NF Rules

- Must be in 2NF.
- Eliminate transitive dependencies (non-key attributes should not depend on other non-key attributes).

3NF Analysis & Adjustments

- **All tables after 2NF**: No transitive dependencies remain, as all non-key attributes depend only on the primary key.

Final 3NF Table Structure:

Table	Primary Key	Key Attributes (Final)
Patient	PatientID	FullName, Gender, Phone, Email, City
Doctor	DoctorID	FullName, Specialty, Phone, Email, ClinicID
Clinic	ClinicID	ClinicName, City, Address
Appointment	AppointmentID	PatientID, DoctorID, ClinicID, Date, StartTime, EndTime, Status
Prescription	PrescriptionID	AppointmentID, MedicineID, IssueDate
Medicine	MedicineID	MedicineName, Type, Description
Inventory	InventoryID	ClinicID, MedicineID, Quantity, ExpiryDate
Billing	BillID	AppointmentID, Amount, BillDate
User	UserID	Username, Password, RoleID
Role	RoleID	RoleName, RoleDescription

Final Relational Schema:

Entity Relationships

- **Patient** (1) — (N) **Appointment**
- **Doctor** (1) — (N) **Appointment**
- **Clinic** (1) — (N) **Doctor**
- **Appointment** (1) — (1) **Prescription**
- **Clinic** (1) — (N) **Inventory**
- **Medicine** (1) — (N) **Inventory**
- **Appointment** (1) — (1) **Billing**
- **User** (N) — (1) **Role**

Roles Table

```
CREATE TABLE Role (
    RoleID INT PRIMARY KEY,
    RoleName VARCHAR(50),
    RoleDescription VARCHAR(200)
);
INSERT INTO Role (RoleID, RoleName, RoleDescription) VALUES
(1, 'Admin', 'Has full access to all system '),
(2, 'Doctor', 'Can manage patients, appointments, and prescriptions'),
(3, 'Receptionist', 'Can manage appointments'),
(4, 'Pharmacist', 'Can manage medicine inventory and prescriptions'),
(5, 'Patient', 'Can view own appointments and medical records');
```

Users Table

```
CREATE TABLE UserAccount (
    UserID INT PRIMARY KEY,
    Username VARCHAR(50) UNIQUE,
    Password VARCHAR(200),
    RoleID INT,
```

```
    FOREIGN KEY (RoleID) REFERENCES Role(RoleID)
);
INSERT INTO UserAccount (UserID, Username, Password, RoleID) VALUES
(1, 'admin1', 'adminpass', 1),
(2, 'admin2', 'pass123', 2),
(3, 'adman3', 'pass123', 2),
(4, 'adman4', 'receppass', 3),
(5, 'user5', 'pharmapass', 4),
(6, 'user6', 'patientpass', 5),
(7, 'user7', 'patientpass', 5),
(8, 'user8', 'pass123', 2),
(9, 'user9', 'receppass', 3),
(10, 'user10', 'patientpass', 5);
```

Clinics Table

```
CREATE TABLE Clinic (
    ClinicID INT PRIMARY KEY,
    ClinicName VARCHAR(100),
    City VARCHAR(50),
    Address VARCHAR(255)
```

```
);

INSERT INTO Clinic (ClinicID, ClinicName, City, Address) VALUES

(1, 'Riyadh Central Clinic', 'Riyadh', '10 King Fahd St'),

(2, 'Riyadh Eastside Clinic', 'Riyadh', '22 Olaya St'),

(3, 'Riyadh Westside Clinic', 'Riyadh', '35 Prince Sultan St'),

(4, 'Jeddah Downtown Clinic', 'Jeddah', '15 Tahlia St'),

(5, 'Jeddah North Clinic', 'Jeddah', '28 Prince Majed St'),

(6, 'Jeddah South Clinic', 'Jeddah', '40 King Abdulaziz St'),

(7, 'Al Khobar Central Clinic', 'Al Khobar', '5 Prince Turki St'),

(8, 'Al Khobar East Clinic', 'Al Khobar', '12 King Fahd Rd'),

(9, 'Al Khobar West Clinic', 'Al Khobar', '20 Prince Nayef St'),

(10, 'Al Khobar Uptown Clinic', 'Al Khobar', '30 King Abdullah St');
```

Patients Table

```
CREATE TABLE Patient (

    PatientID INT PRIMARY KEY,

    FullName VARCHAR(100),

    Gender VARCHAR(10),

    Phone VARCHAR(20),
```

```
    Email VARCHAR(100),
    City VARCHAR(50)
);
INSERT INTO Patient (PatientID, FullName, Gender, Phone, Email, City) VALUES
(1, 'Reem Ahmed', 'Female', '5555', 'reem@example.com', 'Riyadh'),
(2, 'Huda Saad', 'Female', '5551', 'huda@example.com', 'Riyadh'),
(3, 'Lama Khalid', 'Female', '5552', 'lama@example.com', 'Al Khobar'),
(4, 'Ahmed Fares', 'Male', '5553', 'ahmed@example.com', 'Al Khobar'),
(5, 'Abdullah Mohammed', 'Male', '5554', 'abdullah@example.com', 'Al Khobar'),
(6, 'Hadi Fahad', 'Male', '5556', 'hadi@example.com', 'Jeddah'),
(7, 'Sara Mohammed', 'Female', '5557', 'sara@example.com', 'Jeddah'),
(8, 'Maryam Yousif', 'Female', '5558', 'maryam@example.com', 'Al Khobar'),
(9, 'Dima Abdullah', 'Female', '5559', 'dima@example.com', 'Al Khobar'),
(10, 'Noura Abdulrahman', 'Female', '5510', 'noura@example.com', 'Al Khobar');
```

Doctors Table

```
CREATE TABLE Doctor (
    DoctorID INT PRIMARY KEY,
    FullName VARCHAR(100),
    Specialty VARCHAR(100),
```

```
    Phone VARCHAR(20),

    Email VARCHAR(100),

    ClinicID INT,

    FOREIGN KEY (ClinicID) REFERENCES Clinic(ClinicID)

);

INSERT INTO Doctor (DoctorID, FullName, Specialty, Phone, Email, ClinicID)
VALUES

(101, 'Dr. Khaled Mansour', 'Cardiology', '600000001', 'khaled@example.com',
1),

(102, 'Dr. Lina Luay', 'Dermatology', '600000002', 'lina@example.com', 2),

(103, 'Dr. Faisal Salah', 'Pediatrics', '600000003', 'faisal@example.com', 3),

(104, 'Dr. Rana Fahad', 'Orthopedics', '600000004', 'rana@example.com', 4),

(105, 'Dr. Majed Rashed', 'Neurology', '600000005', 'majed@example.com', 5),

(106, 'Dr. Sara Omar ', 'General Medicine', '600000006', 'sara@example.com', 6),

(107, 'Dr. Salim Saad', 'ENT', '600000007', 'salim@example.com', 7),

(108, 'Dr. Huda Omar', 'Ophthalmology', '600000008', 'huda@example.com', 8),

(109, 'Dr. Omar Mourad', 'Gastroenterology', '600000009',
'omar@example.com', 9),

(110, 'Dr. Maha Yaseer', 'Cardiology', '600000010', 'maha@example.com', 10);
```

Appointments Table

```
CREATE TABLE Appointment (
    AppointmentID INT PRIMARY KEY,
    PatientID INT,
    DoctorID INT,
    ClinicID INT,
    AppointmentDate DATE,
    StartTime TIME,
    EndTime TIME,
    Status VARCHAR(50),
    FOREIGN KEY (PatientID) REFERENCES Patient(PatientID),
    FOREIGN KEY (DoctorID) REFERENCES Doctor(DoctorID),
    FOREIGN KEY (ClinicID) REFERENCES Clinic(ClinicID)
);
INSERT INTO Appointment (AppointmentID, PatientID, DoctorID, ClinicID,
AppointmentDate, StartTime, EndTime, Status) VALUES
(1001, 1, 101, 1, '2025-12-01', '08:30', '09:00', 'Scheduled'),
(1002, 2, 102, 2, '2025-12-02', '09:15', '09:45', 'Completed'),
(1003, 3, 103, 3, '2025-12-03', '10:00', '10:30', 'Scheduled'),
(1004, 4, 104, 4, '2025-12-04', '11:00', '11:30', 'Cancelled'),
```

```
(1005, 5, 105, 5, '2025-12-05', '12:30', '13:00', 'Scheduled'),
(1006, 6, 106, 6, '2025-12-06', '13:15', '13:45', 'Scheduled'),
(1007, 7, 107, 7, '2025-12-07', '14:00', '14:30', 'Completed'),
(1008, 8, 108, 8, '2025-12-08', '15:00', '15:30', 'Scheduled'),
(1009, 9, 109, 9, '2025-12-09', '16:00', '16:30', 'Scheduled'),
(1010, 10, 110, 10, '2025-12-10', '17:00', '17:30', 'Scheduled');
```

Medicines Table

```
CREATE TABLE Medicine (
    MedicineID INT PRIMARY KEY,
    MedicineName VARCHAR(100),
    Type VARCHAR(50),
    Description VARCHAR(255)
);
INSERT INTO Medicine (MedicineID, MedicineName, Type, Description) VALUES
(211, 'Vitamin C', 'Tablet', 'Boosts immunity'),
(212, 'Buscopan', 'Tablet', 'Relieves abdominal pain'),
(213, 'Panadol', 'Tablet', 'Pain reliever & fever reducer'),
(214, 'Aspirin', 'Tablet', 'Pain reliever'),
(215, 'Omeprazole', 'Capsule', 'Reduces stomach acid'),
```

(216, 'Amoxicillin', 'Capsule', 'Antibiotic'),

(217, 'Artelac', 'Eye Drops', 'Moisturizes eyes'),

(218, 'Duspatalin', 'Capsule', 'Relieves gastrointestinal pain'),

(219, 'Stfal Cream', 'Cream', 'Moisturizes facial skin'),

(220, 'Cetirizine', 'Tablet', 'Relieves allergy symptoms');

Prescriptions Table

```
CREATE TABLE Prescription (
    PrescriptionID INT PRIMARY KEY,
    AppointmentID INT,
    MedicineID INT,
    IssueDate DATE,
    FOREIGN KEY (AppointmentID) REFERENCES Appointment(AppointmentID),
    FOREIGN KEY (MedicineID) REFERENCES Medicine(MedicineID)
);
```

INSERT INTO Prescription (PrescriptionID, AppointmentID, MedicineID, IssueDate) VALUES

(2001, 1001, 211, '2025-12-01'),

```
(2002, 1002, 212, '2025-12-02'),
(2003, 1003, 213, '2025-12-03'),
(2004, 1004, 214, '2025-12-04'),
(2005, 1005, 215, '2025-12-05'),
(2006, 1006, 216, '2025-12-06'),
(2007, 1007, 217, '2025-12-07'),
(2008, 1008, 218, '2025-12-08'),
(2009, 1009, 219, '2025-12-09'),
(2010, 1010, 220, '2025-12-10');
```

Inventory Table

```
CREATE TABLE Inventory (
    InventoryID INT PRIMARY KEY,
    ClinicID INT,
    MedicineID INT,
    Quantity INT,
    ExpiryDate DATE,
    FOREIGN KEY (ClinicID) REFERENCES Clinic(ClinicID),
    FOREIGN KEY (MedicineID) REFERENCES Medicine(MedicineID)
```

```
);


INSERT INTO Inventory (InventoryID, ClinicID, MedicineID, Quantity, ExpiryDate)
VALUES

(1001, 1, 211, 30, '2026-12-20'),

(1002, 2, 212, 25, '2026-11-20'),

(1003, 3, 213, 40, '2026-11-20'),

(1004, 4, 214, 15, '2026-11-20'),

(1005, 5, 215, 60, '2026-11-20'),

(1006, 6, 216, 35, '2026-11-20'),

(1007, 7, 217, 50, '2026-11-20'),

(1008, 8, 218, 20, '2026-11-20'),

(1009, 9, 219, 45, '2026-11-20'),

(1010, 10, 220, 30, '2026-11-20');
```

Billing Table

```
CREATE TABLE Billing (

    BillID INT PRIMARY KEY,

    AppointmentID INT,

    Amount DECIMAL(10,2),
```

```
    BillDate DATE,
    FOREIGN KEY (AppointmentID) REFERENCES Appointment(AppointmentID)
);

INSERT INTO Billing (BillID, AppointmentID, Amount, BillDate) VALUES
(4001, 1001, 150.00, '2025-11-20'),
(4002, 1002, 200.00, '2025-11-21'),
(4003, 1003, 180.00, '2025-11-22'),
(4004, 1004, 250.00, '2025-11-23'),
(4005, 1005, 160.00, '2025-11-24'),
(4006, 1006, 220.00, '2025-11-25'),
(4007, 1007, 190.00, '2025-11-26'),
(4008, 1008, 210.00, '2025-11-27'),
(4009, 1009, 230.00, '2025-11-28'),
(4010, 1010, 170.00, '2025-11-29');
```

Queries

Patients with more than one appointment

```
SELECT p.FullName, COUNT(a.AppointmentID) AS NumAppointments
```

```
FROM Patient p
JOIN Appointment a ON p.PatientID = a.PatientID
GROUP BY p.FullName
HAVING COUNT(a.AppointmentID) > 1;
```

Doctors by specialty

```
SELECT Specialty, COUNT(DoctorID) AS NumDoctors
FROM Doctor
GROUP BY Specialty;
```

Medicines with low stock < 20

```
SELECT m.MedicineName, i.Quantity, c.ClinicName
FROM Inventory i
JOIN Medicine m ON i.MedicineID = m.MedicineID
JOIN Clinic c ON i.ClinicID = c.ClinicID
WHERE i.Quantity < 20;
```

appointments

```
SELECT a.AppointmentID, p.FullName AS Patient, d.FullName AS Doctor,
a.AppointmentDate

FROM Appointment a

JOIN Patient p ON a.PatientID = p.PatientID

JOIN Doctor d ON a.DoctorID = d.DoctorID

WHERE a.AppointmentDate >= CURRENT_DATE;
```

Authentication

```
SELECT * FROM UserAccount WHERE Username='admin1' AND
Password='adminpass';
```

Encryption

```
INSERT INTO UserAccount (UserID, Username, Password, RoleID)

VALUES (11, 'newuser', SHA2('mypassword', 256), 5);
```

There is note at the end

Available Tables

Appointment

AppointmentID	PatientID	DoctorID	Clinic
1001	1	101	1
1002	2	102	2
1003	3	103	3
1004	4	104	4
1005	5	105	5
1006	6	106	6
1007	7	107	7
1008	8	108	8
1009	9	109	9
1010	10	110	10

Appointment

AppointmentDate	StartTime	EndTime	Stat
2025-12-01	08:30	09:00	Sch
2025-12-02	09:15	09:45	Con
2025-12-03	10:00	10:30	Sch
2025-12-04	11:00	11:30	Can
2025-12-05	12:30	13:00	Sch
2025-12-06	13:15	13:45	Sch
2025-12-07	14:00	14:30	Con
2025-12-08	15:00	15:30	Sch
2025-12-09	16:00	16:30	Sch
2025-12-10	17:00	17:30	Sch

Billing

BillID	AppointmentID	Amount	BillDate
4001	1001	150	2025-11-20
4002	1002	200	2025-11-21
4003	1003	180	2025-11-22
4004	1004	250	2025-11-23
4005	1005	160	2025-11-24
4006	1006	220	2025-11-25
4007	1007	190	2025-11-26
4008	1008	210	2025-11-27
4009	1009	230	2025-11-

Clinic

ClinicID	ClinicName	City	Address
1	Riyadh Central Clinic	Riyadh	10 King Fahd St
2	Riyadh Eastside Clinic	Riyadh	22 Olaya St
3	Riyadh Westside Clinic	Riyadh	35 Prince Sultan St
4	Jeddah Downtown Clinic	Jeddah	15 Tahlia St
5	Jeddah North Clinic	Jeddah	28 Prince Majed St
6	Jeddah South Clinic	Jeddah	40 King Abdulaziz St
7	Al Khobar Central Clinic	Al Khobar	5 Prince Turki St
8	Al Khobar East Clinic	Al Khobar	12 King Fahd Rd
9	Al Khobar West Clinic	Al Khobar	20 Prince Nayef St
10	Al Khobar Uptown Clinic	Al Khobar	30 King Abdullah St

Doctor

DoctorID	FullName	Specialty	Ph
101	Dr. Khaled Mansour	Cardiology	60
102	Dr. Lina Luay	Dermatology	60
103	Dr. Faisal Salah	Pediatrics	60
104	Dr. Rana Fahad	Orthopedics	60
105	Dr. Majed Rashed	Neurology	60
106	Dr. Sara Omar	General Medicine	60
107	Dr. Salim Saad	ENT	60
108	Dr. Huda Omar	Ophthalmology	60

Doctor

Phone	Email	ClinicID
600000001	khaled@example.com	1
600000002	lina@example.com	2
600000003	faisal@example.com	3
600000004	rana@example.com	4
600000005	majed@example.com	5
600000006	sara@example.com	6
600000007	salim@example.com	7
600000008	huda@example.com	8

Inventory

InventoryID	ClinicID	MedicineID	Quanti
1001	1	211	30
1002	2	212	25
1003	3	213	40
1004	4	214	15
1005	5	215	60
1006	6	216	35
1007	7	217	50
1008	8	218	20
1009	9	219	45
1010	10	220	30

Inventory

linicID	MedicineID	Quantity	ExpiryDate
	211	30	2026-12-20
	212	25	2026-11-20
	213	40	2026-11-20
	214	15	2026-11-20
	215	60	2026-11-20
	216	35	2026-11-20
	217	50	2026-11-20
	218	20	2026-11-20
	219	45	2026-11-20
)	220	30	2026-11-20

Medicine

MedicineID	MedicineName	Type	Des
211	Vitamin C	Tablet	Boo imr
212	Buscopan	Tablet	Rel abo
213	Panadol	Tablet	Pai fev
214	Aspirin	Tablet	Pai
215	Omeprazole	Capsule	Rec sto
216	Amoxicillin	Capsule	Ant
217	Artelac	Eye Drops	Mo eye
218	Duspatalin	Capsule	Rel gas pai
219	Stfal Cream	Cream	Mo fac
			Rel

Medicine

MedicineName	Type	Description
Vitamin C	Tablet	Boosts immunity
Buscopan	Tablet	Relieves abdominal pain
Panadol	Tablet	Pain reliever & fever reducer
Aspirin	Tablet	Pain reliever
Omeprazole	Capsule	Reduces stomach acid
Amoxicillin	Capsule	Antibiotic
Artelac	Eye Drops	Moisturizes eyes
Duspatalin	Capsule	Relieves gastrointestinal pain
Stfal Cream	Cream	Moisturizes facial skin
		Relieves alleray

Patient

PatientID	FullName	Gender	Phone
1	Reem Ahmed	Female	5555
2	Huda Saad	Female	5551
3	Lama Khalid	Female	5552
4	Ahmed Fares	Male	5553
5	Abdullah Mohammed	Male	5554
6	Hadi Fahad	Male	5556
7	Sara Mohammed	Female	5557
8	Maryam Yousif	Female	5558
9	Dima Abdullah	Female	5559
10	Noura Abdulrahman	Female	5510

Patient

Phone	Email	City
5555	reem@example.com	Riyadh
5551	huda@example.com	Riyadh
5552	lama@example.com	Al Khobar
5553	ahmed@example.com	Al Khobar
5554	abdullah@example.com	Al Khobar
5556	hadi@example.com	Jeddah
5557	sara@example.com	Jeddah
5558	maryam@example.com	Al Khobar
5559	dima@example.com	Al Khobar
5510	noura@example.com	Al Khobar

Prescription

PrescriptionID	AppointmentID	MedicineID
2001	1001	211
2002	1002	212
2003	1003	213
2004	1004	214
2005	1005	215
2006	1006	216
2007	1007	217
2008	1008	218
2009	1009	219

Prescription

D	AppointmentID	MedicineID	IssueDate
	1001	211	2025-12-01
	1002	212	2025-12-02
	1003	213	2025-12-03
	1004	214	2025-12-04
	1005	215	2025-12-05
	1006	216	2025-12-06
	1007	217	2025-12-07
	1008	218	2025-12-08
	1009	219	2025-12-09

Role

RoleID	RoleName	RoleDescription
1	Admin	Has full access to all system
2	Doctor	Can manage patients, appointments, and prescriptions
3	Receptionist	Can manage appointments
4	Pharmacist	Can manage medicine inventory and prescriptions
5	Patient	Can view own appointments and medical records

UserAccount

UserID	Username	Password	RoleID
1	admin1	adminpass	1
2	admin2	pass123	2
3	adman3	pass123	2
4	adman4	receppass	3
5	user5	pharmapass	4
6	user6	patientpass	5
7	user7	patientpass	5
8	user8	pass123	2
9	user9	receppass	3
10	user10	patientpass	5

Output

Specialty	NumDoctors
Cardiology	2
Dermatology	1
ENT	1
Gastroenterology	1
General Medicine	1
Neurology	1
Ophthalmology	1
Orthopedics	1
Pediatrics	1

MedicineName	Quantity	ClinicName
Aspirin	15	Jeddah Downtown Clinic

AppointmentID	Patient	Doctor	AppointmentDate
1001	Reem Ahmed	Dr. Khaled Mansour	2025-12-01
1002	Huda Saad	Dr. Lina Luay	2025-12-02
1003	Lama Khalid	Dr. Faisal Salah	2025-12-03
1004	Ahmed Fares	Dr. Rana Fahad	2025-12-04
1005	Abdullah Mohammed	Dr. Majed Rashed	2025-12-05
1006	Hadi Fahad	Dr. Sara Omar	2025-12-06
1007	Sara Mohammed	Dr. Salim Saad	2025-12-07
1008	Maryam Yousif	Dr. Huda Omar	2025-12-08
1009	Dima Abdullah	Dr. Omar Mourad	2025-12-09
1010	Noura Abdulrahman	Dr. Maha Yaseer	2025-12-10

UserID	Username	Password	RoleID
1	admin1	adminpass	1

Notes:

I am Using SQLite Online, so SHA2 function is not supported.

It should be:

INSERT INTO UserAccount (UserID, Username, Password, RoleID)

VALUES (11, 'newuser', SHA2('mypassword', 256), 5);

But I wrote it in SQLite Online:

INSERT INTO UserAccount (UserID, Username, Password, RoleID)

VALUES (11, 'newuser', 'mypassword', 5);